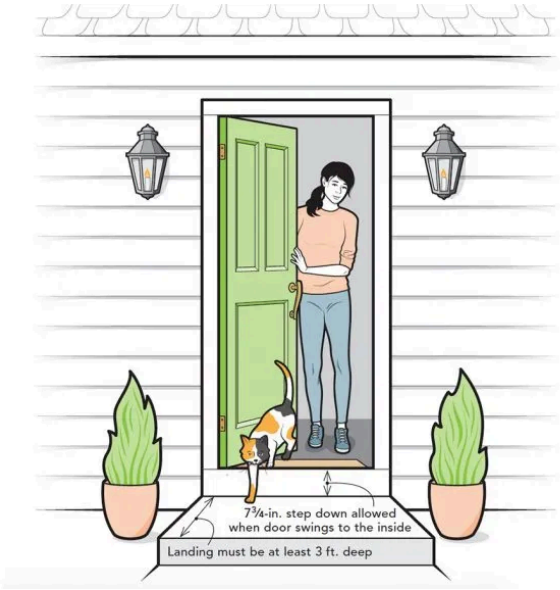
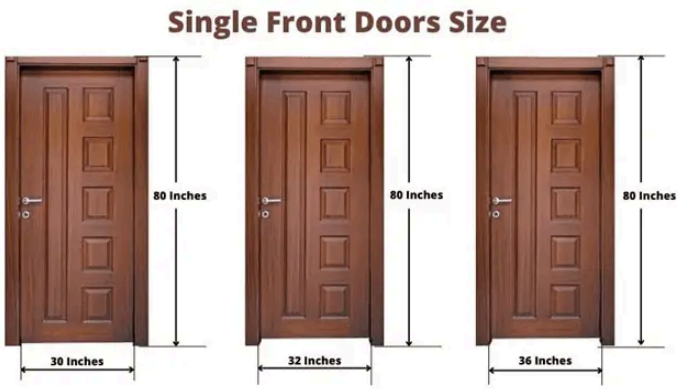
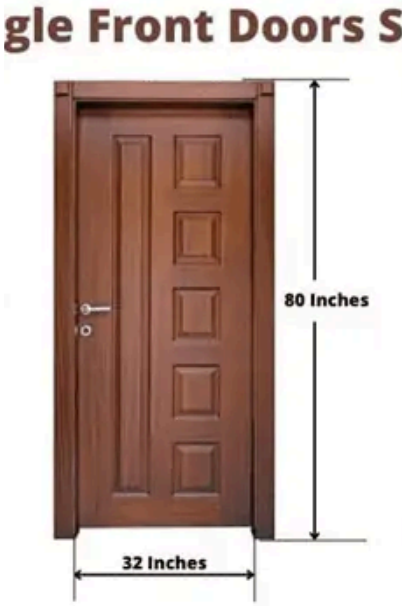
Course Title: Exterior Door Compliance	**Audience:** Construction Manager
Needs Analysis ● **Learners:** Construction Managers overseeing residential projects, particularly those working with single-family homes or similar developments. ● **Experience Level**: Mid-level or experienced professionals who may already have a basic understanding of construction but need to refine their knowledge of International Residential Code (IRC) code compliance. ● **Sticking Points**: Ensuring designs meet multiple regulatory requirements (e.g., accessibility, fire safety, weather resistance, structural integrity); Managing multiple vendors and contractors to ensure compliance with codes. ● **KPIs**: Course completion; Percentage of employers reporting noticeable improvements in employees using IRC to support decision-making.	
Course Description: This dynamic course is designed to equip you with the critical skills needed to navigate and apply the International Residential Code (IRC). You will explore real-world scenarios where your decisions directly impact the safety, accessibility, and structural integrity of homes.	**Learning Objectives:** After completing this course, learners will be able to: ● Apply IRC guidelines to select the correct materials, flashing, and configurations for exterior doors in residential construction. ● Analyze exterior door designs to determine compliance with IRC structural, safety, and accessibility standards.
Assessment Type: Gamified Challenge: Correct answers throughout the questions in the challenge receive points that total up to a final score.	

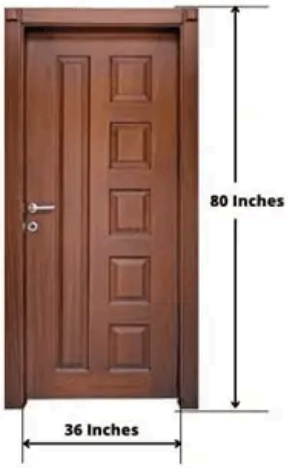
Page	Page Text	Design
1	**Page Title:** Welcome to Brookstone Meadows **Page Content:** Welcome, [First Name]! You've the Construction Manager for the Brookstone Meadows single-family home development project. Brookstone Meadows expects all your decisions to be cost-effective and comply with the structural, safety, and accessibility provisions outlined in the **International Residential Code (IRC)**. Your first major task is to ensure that all **exterior doors** in the new homes comply with the **structural, safety, and accessibility** provisions in the IRC. Throughout this course, you'll face challenges and must make informed decisions. Click the **IRC icon** on	**Page style:** “Introduction” template page for construction series **Design Elements:** ● Emphasize the **IRC Icon** with a tooltip or pop-up to demo how to open the sections of the IRC. ● Include a brief example question along with the scoring system, showing how points are awarded for each correct answer.

	each page to review the applicable standards before making your final decisions. Your goal is to accumulate **60 points or more** by making the most effective decisions. This will demonstrate your ability to successfully manage construction requirements and maintain high-quality standards for the Brookstone Meadows development. Are you ready?	
2	**Page Title:** Exterior Door Landing Design **Page Content:** You begin by inspecting the door landing designs that were submitted last week. Before approving the designs, check **IRC Section R311.3.2**, which outlines the landing and threshold height requirements. Then, determine if each design meets the code by selecting **Compliant** or **Non-compliant**. **A. Compliant** **Front Door Design**: This design features a **single step** with a step-down height of **6 inches**. *NOTE: These are example illustrations (Source)*  **B. Compliant** **Back Door Design**: This design features a **double**	**Page style:** "Question with images" template page for construction series **Design Elements (Examples):** ● Media: IMG_IRC1 ● Media: IMG_IRC2 ● Score - Show points earned **Interaction**: ● The **IRC Icon** opens **Section R311.3.2**. ● Use **interactive buttons** for selecting answers. ● After click, show **explanation** beneath each image. ● Include **images (like IMG_IRC1 and IMG_IRC2)** of the door designs to illustrate the different step/landing configurations.

	step with a total stair height of **10 inches**. **Explanation**: ● **Front Door**: The 6-inch step-down height is compliant with the **IRC**, which permits a **maximum step height of 7 ¾ inches**. ● **Back Door**: The total rise of 10 inches (for two steps) is compliant with the **IRC**, which allows for a **maximum total rise of 15 ½ inches** for steps.	
3	**Page Title:** Exterior Door Materials **Page Content:** Next, you have a meeting with suppliers to evaluate and select the best material for the external doors based on **structural integrity** and **weather resistance**. Before you make a decision, review **Section R609** of the IRC. Then, select one type of door to recommend for Brookstone Meadows homes. A. **Fiberglass door** with a **solid core** and **insulated glazing** B. **Hollow-core metal door** with a **steel frame** and no insulation C. **Vinyl door** with **hollow frame** and no thermal	**Page style:** "Question (Multiple Choice)" template page for construction series **Design Elements (Examples):** ● Score - Show points earned **Interaction**: ● The **IRC Icon** opens **Section R609**. ● Use **interactive buttons** for selecting answers. Randomize answer choices.

	barrier **Explanation:** The **fiberglass door with a solid core** and **insulated glazing** is compliant with the **IRC** code. Fiberglass doors are **strong**, **durable**, and offer excellent **insulation**, which is critical for energy efficiency and protection against the elements. The **insulated glazing** helps reduce heat transfer and condensation, ensuring the door maintains its **thermal performance**. The IRC requires exterior doors to be properly insulated to prevent moisture infiltration, heat loss, and provide long-term durability.	● After click, show **explanation**.
4	**Page Title:** Egress Door Width **Page Content:** During your meeting, the suppliers also ask you to evaluate the size of the egress doors for the houses with basements. **IRC Section R311.2** outlines the accessibility codes for residential doors. Determine if each design meets the code by selecting **Compliant** or **Non-compliant**. *NOTE: This is an example illustration (Source)*  **A. Non-compliant**	**Page style:** "Question with images" template page for construction series **Design Elements (Examples):** ● Media: IMG_IRC3 ● Score - Show points earned **Interaction**: ● The **IRC Icon** opens **Section R311.2**. ● Use **interactive buttons** for selecting answers. ● After click, show **explanation** beneath each image. ● Include **images (like IMG_IRC3)** of the door designs to illustrate the sizes of the door.

	 B. Compliant  **C. Compliant**	

	Size  **Explanation:** **Door 1** is **non-compliant** with **IRC accessibility** standards, which requires doors to be at least **32 inches** wide. **Door 2** is **compliant** with **IRC accessibility** standards. It meets the required **clearance** and **threshold height** for safe and easy access, especially for individuals with disabilities. **Door 3** is **compliant** with **IRC accessibility** standards. It meets the required **clearance** and **threshold height** for safe and easy access, especially for individuals with disabilities.	
5	**Page Title:** External Door and Fire Safety **Page Content:** A few weeks later, the doors you selected are installed in several homes. You drive to the construction site to check on the progress, focusing on whether the doors meet the **fire safety** requirements outlined in the **IRC**. Review **Sections R609.1** and **R703.4**, and determine if each design meets the code by selecting **Compliant** or **Non-compliant**. *Example Images (Source)* **A. Compliant**	**Page style:** "Question with images" template page for construction series **Design Elements (Examples):** ● Media: IMG_IRC4 ● Media: IMG_IRC5 ● Media: IMG_IRC6 ● Score - Show points earned **Interaction**: ● The **IRC Icon** opens **Sections R609.1** and

	  B. Non-compliant	**R703.4**. ● Use **interactive buttons** for selecting answers. ● After click, show **explanation** beneath each image. ● Include **images (like IMG_IRC4, IMG_IRC5, IMG_IRC6)** of the door designs to illustrate flashing.

	  C. Non-compliant	

	   Explanation: **Door 1** is **compliant** because this door frame includes the **proper flashing** installation as outlined in **Section R703.4**. Flashing is crucial in directing water away from door frames and preventing moisture from deteriorating the door's fire-resistant properties. A properly flashed door ensures that water does not seep into the door assembly, which can weaken fire-resistant materials or create gaps that compromise the door's performance in a fire. **Door 2** is **non-compliant** because this door frame does not include the **proper flashing**. **Door 3** is **non-compliant** because this door frame does not include the **proper flashing**.	
6	**Page Title:** Conclusion	**Page style:** "Conclusion" template

	Page Content: Congratulations, [First Name]! You've completed the challenge! Here's how you did: Your Final Score: [Insert Score] *(Out of 90 points)* ***If score is 60 or higher*** 🎉 You've demonstrated a solid understanding of the IRC requirements and are ready to apply these skills to your construction projects. **Keep up the great work, and use this knowledge to make smart, compliant decisions on the job!** ***If score is below 60*** ⚠️ While you've gained valuable insights, you'll need to revisit some key concepts to ensure you're fully prepared for real-world application. **We recommend reviewing the material again to strengthen your understanding and make sure you're ready for success in your next project.**	page for construction series **Design Elements:** ● Show final score.